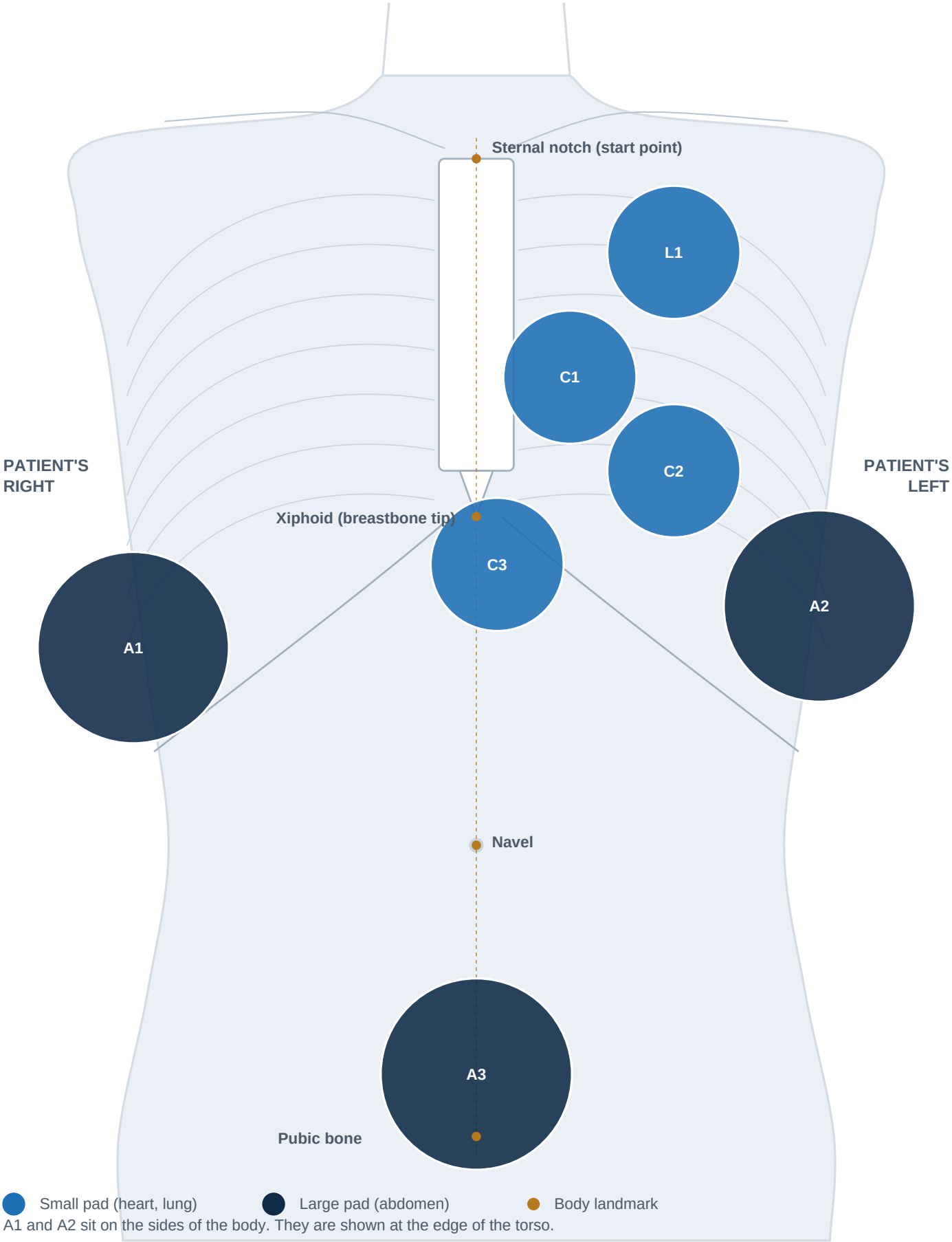
Where each pad goes

Front view of the torso, drawn at 40% size. Patient's left is on the right of the page, like facing the patient.



Small pad (heart, lung) Large pad (abdomen) Body landmark

A1 and A2 sit on the sides of the body. They are shown at the edge of the torso.

Concept drawing. Prototype aid only, not a medical device. Landmarks are for an average adult: check with a trained sonographer or flight surgeon.

Landmark measurements

Measure from the sternal notch (hollow at the base of the throat). Scaled for notch-to-xiphoid = 17.0 cm.

Pad	Window	From notch	Where to feel for it	Probe
C1	Heart: parasternal	4.5 cm left, 10.5 cm down	Left of the breastbone, 4th rib space	Marker toward the right shoulder
C2	Heart: apical	9.5 cm left, 15.0 cm down	Under the left nipple line, 5th rib space (where the beat is felt)	Marker toward the left side
C3	Heart and vena cava: subcostal	1.0 cm left, 19.5 cm down	Just below the breastbone tip, probe aimed up toward the left shoulder	Marker toward the left side
L1	Lung (sliding check)	9.5 cm left, 4.5 cm down	Below the collarbone, 2nd rib space, nipple line. Print two pads for left and right	Marker toward the head
A1	Right flank (liver, kidney)	16.5 cm right, 23.5 cm down	Right side between ribs 8 and 11, on the mid-side line	Marker toward the head
A2	Left flank (spleen, kidney)	16.5 cm left, 21.5 cm down	Left side between ribs 6 and 9, a little behind the mid-side line	Marker toward the head
A3	Pelvis (bladder, free fluid)	Midline, 44.0 cm down	Middle, just above the pubic bone	Marker toward the right side

How to place a pad

1. Clean and dry the skin. Peel the pad from its backing. Do not touch the adhesive side.
2. Find the landmark, measure the distance in the table, and mark the spot with a skin-safe pen.
3. Put the centre of the slot on the spot. Turn the pad so the triangle (marker mark) points the way in the table.
4. Press from the centre outward for 10 seconds. Gel goes inside the slot only, never under the ring.
5. Photograph each placed pad. The app uses the photo to log the pad position.

Check before use

Sample positions only. Body size, microgravity posture and the probe model change the best spot.
A trained sonographer or flight surgeon must confirm every window before crew use.

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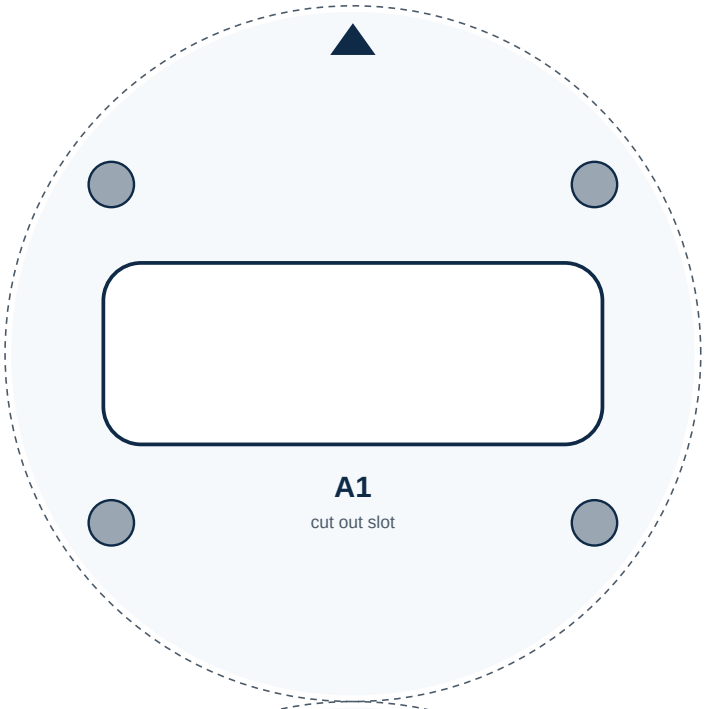
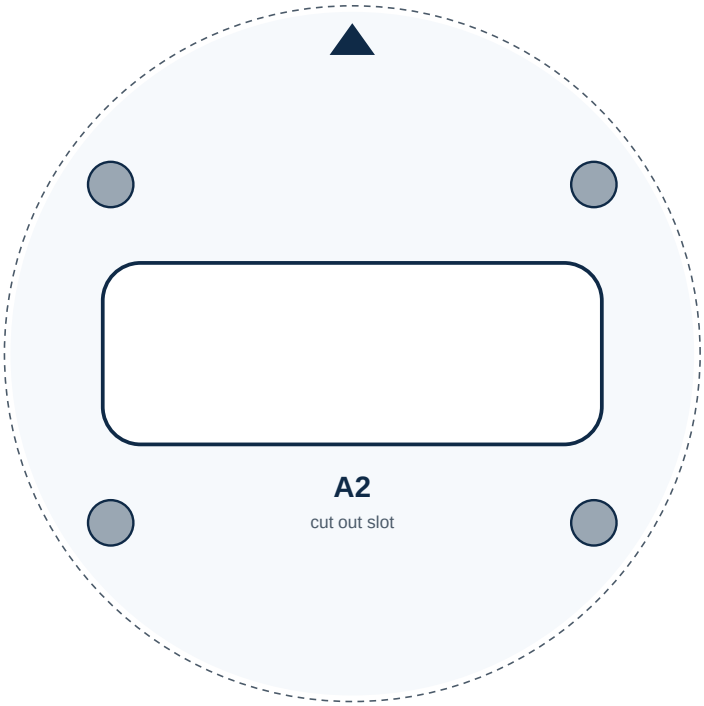
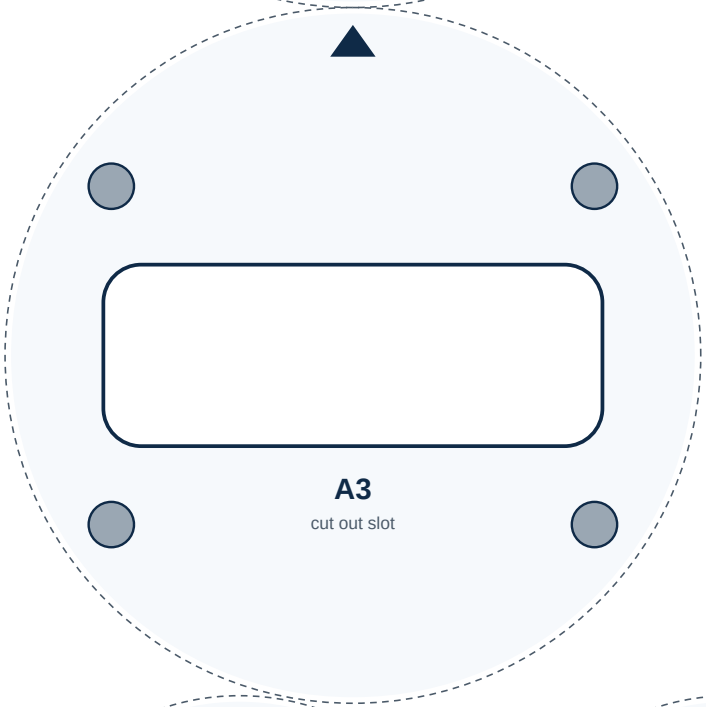
Pads at actual size (100%)

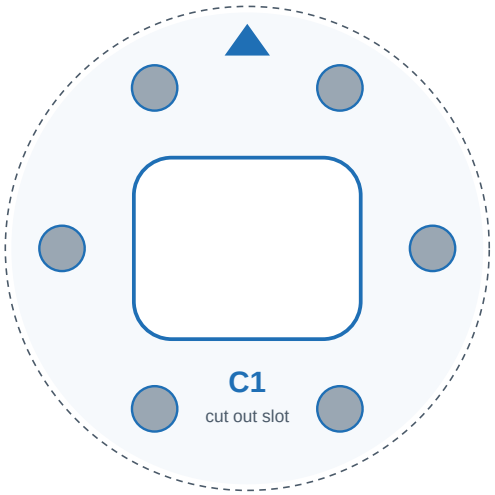
This bar must measure 50 mm

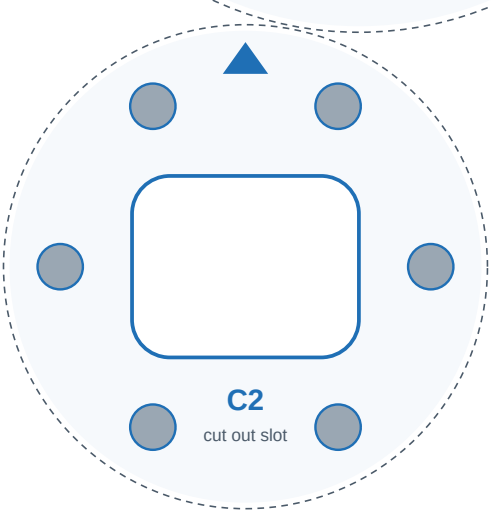
Print with "Actual size" or "100%", no fit to page. Check the 50 mm bar (top right) before cutting.

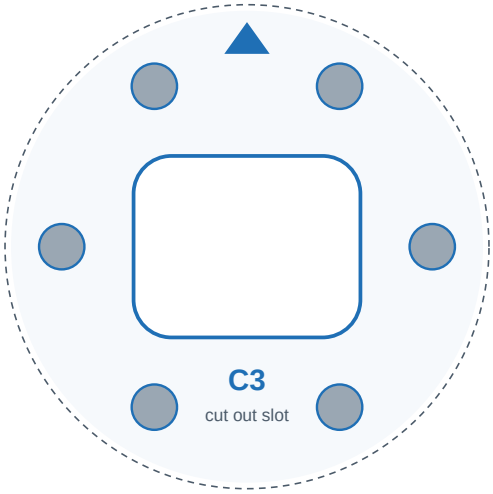


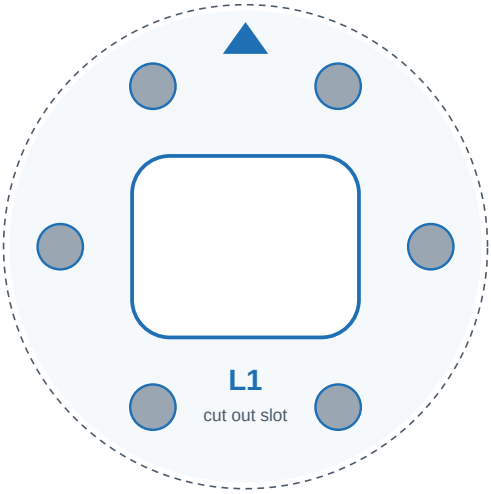












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Prototype build sheet

A bench test for the idea: magnetic collar on the probe, steel-dot pad on a body model. Not for use on people yet.

Materials (about one hour, low cost)

- This printed sheet on adhesive label paper or clear film. Laminate or seal with clear tape for gel resistance.
- Flat steel discs, 6 mm wide and 0.5 to 1 mm thick: 6 for each small pad, 4 for each large pad (washers or cut shim steel).
- Small neodymium magnets, 6 mm wide and 2 mm thick, grade N35 (weak ones): 6 or 4 for each probe collar, to match the pad.
- A 3D-printed or cardboard collar that clips around your probe or a dummy probe, with magnet pockets on the same circle as the dots.
- Hypoallergenic medical tape for the pad edge. Ultrasound gel. A skin-safe marker.
- A body model: a gel phantom, a foam torso, or a bag of gel on a board. Use a phantom first.

Build steps

1. Print page 3 at 100%. Check the 50 mm bar. Cut the outer circle and the slot.
2. Glue one steel disc on each grey circle, under the label film, so the surface stays smooth.
3. Make the collar so the probe face sits flush with the slot, and the pockets line up with the dots when the collar marker triangle faces the pad triangle.
4. Press a magnet into each pocket. Use the same pole facing out on all of them, and mark it. Never let two magnets snap together on a finger.
5. Stick a pad on the phantom. Lower the collar. It should seat itself and keep the probe aimed without a hand.
6. Add light pressure with a spring or soft foam inside the collar, so the probe keeps skin contact as the body moves.

Tests to run and record (good evidence for the submission)

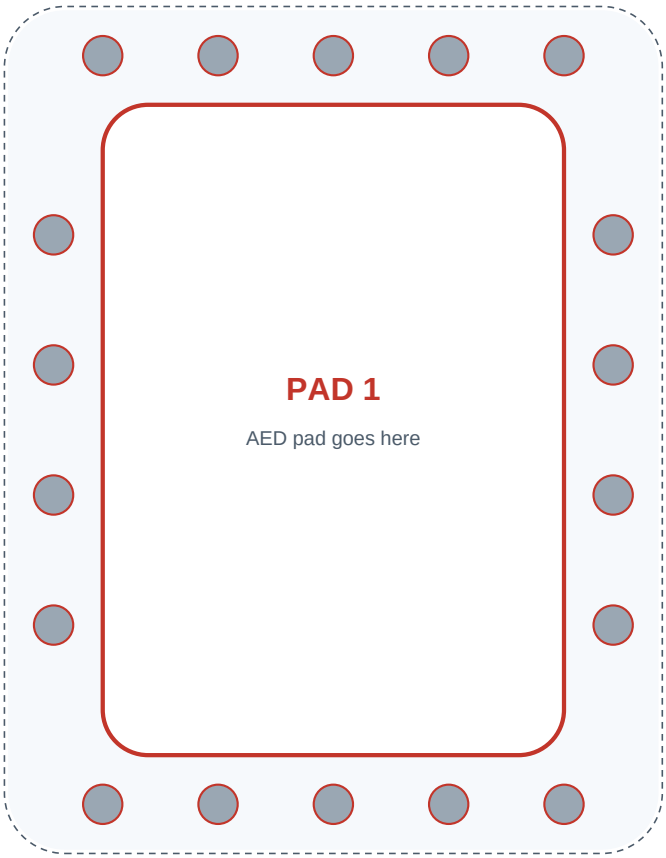
- Pull-off force with a kitchen or luggage scale, collar to pad, straight up and sideways. Record N.
- Does the image stay in the same place for 60 seconds with no hand? Record a screen video.
- Does the magnet change the image or add noise? Compare 10 seconds with the collar on and off.
- Do 5 people who are not doctors find the right window in the app using only the on-screen guide? Time them.

Safety

Do not put magnets near pacemakers, defibrillators, insulin pumps or other implants. Keep them away from anyone with such a device.
Start on a phantom. Use weak magnets. Keep strong magnets apart from each other and from steel tools, they can pinch hard.
Magnets hold the probe sideways and in angle. They do not set contact pressure; a spring or the user does.
A magnetic hold is a convenience, not a safety system. Always have a quick manual release.
Magnets do not harm ultrasound pictures in most probes. Measure it, as test 3 above does.

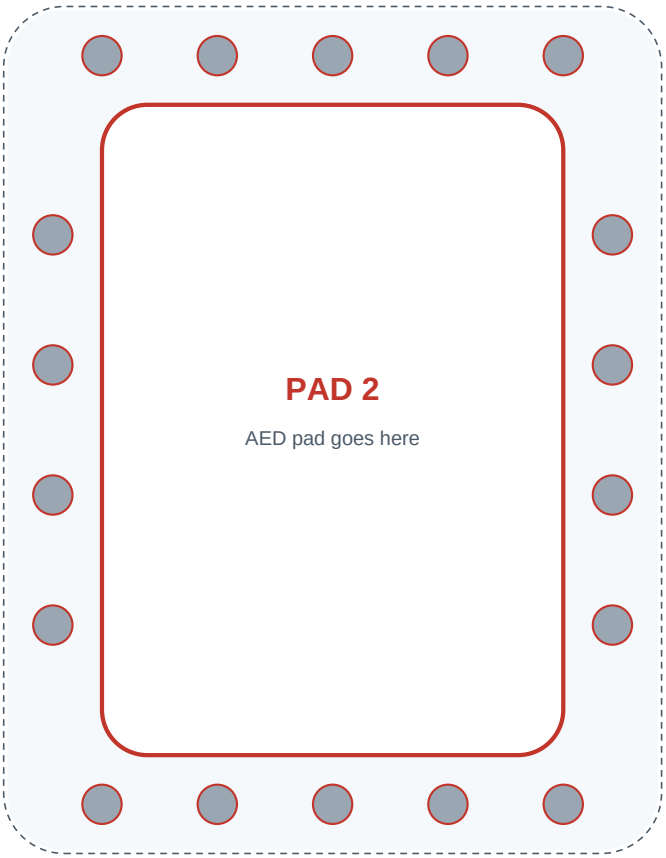
Emergency chest patch (AED dock)

Concept: two magnetic dock frames hold the AED pads in the right place. Print at 100%.



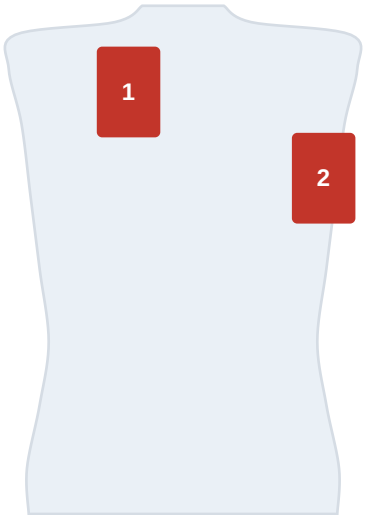
Patient's RIGHT upper chest

Below the collarbone, right of the breastbone



Patient's LEFT side

Below the armpit, on the ribs at the side



How it works (concept)

- Wear or stick the patch before a risk period. Steel dots mark and hold the frame.
- An AED pad has to touch skin. The magnet only holds the pad in place and shows where it goes.
- For adults with a standard AED, pad positions follow the usual layout. Always follow the AED voice prompts.
- Use the pediatric pads and settings if the device offers them. Not tested: this is a prototype aid.

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